

Review

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1 Notes

Sps $f : [a, b] \rightarrow \mathbb{R}$ is bdd by $M > 0$ and f is RI. Show that f^2 is RI.

Fix a partition $a = x_0 < x_1 < x_2 < \cdots < x_n = b$ of $I = [a, b]$. Let $M_i = \sup\{f(x) : x \in [x_{i-1}, x_i]\}$, $m_i = \inf\{f(x) : x \in [x_{i-1}, x_i]\}$ and $M_i^* = \sup\{f(x)^2 : x \in [x_{i-1}, x_i]\}$ and $m_i^* = \inf\{f(x)^2 : x \in [x_{i-1}, x_i]\}$ for $i \in [n]$.

Let $\epsilon > 0$ and $i \in [n]$. There exist $x, y \in [x_{i-1}, x_i]$ such that $f(x)^2 > M_i^* - \epsilon$ and $f(y)^2 < m_i^* + \epsilon$, so

$$\begin{aligned} M_i^* - m_i^* - 2\epsilon &\leq f(x)^2 - f(y)^2 \leq 2M \cdot (f(x) - f(y)) \\ &\leq 2M \cdot (M_i - m_i). \end{aligned}$$

Then

$$pc \int_I \bar{g} - \underline{g} \leq 2\epsilon \cdot (b - a) + 2M \cdot pc \int_I \bar{f} - \underline{f}.$$

Then we can take simultaneously $\epsilon \rightarrow 0$ and $pc \int_I \bar{f} - \underline{f}$ by refining the partition $\{x_i\}$, so $g = f^2$ is RI.

1. sps $F, G : [a, b] \rightarrow \mathbb{R}$ are diff and $f = F', g = G'$ are RI. Show that

$$\int_a^b F(x)g(x)dx + \int_a^b f(x)G(x)dx = F(b)G(b) - F(a)G(a).$$

Proof: Since F, G are diff, F, G are cont, so F, G are RI, so Fg, fG are RI, so $Fg + fG = (FG)'$ is RI.

$(FG)'$ is RI with antiderivative FG , so by the fundamental theorem,

$$\int_a^b (FG)'(x)dx = F(b)G(b) - F(a)G(a),$$

and equals $\int_a^b F(x)g(x)dx + \int_a^b f(x)G(x)dx$.

Sps $\varphi : [a, b] \rightarrow [A, B]$ with $a < b$, $\varphi'(x) > 0$ is continuous, $\varphi(a) = A$, $\varphi(b) = B$ and $f : [A, B] \rightarrow \mathbb{R}$ is cont. Then

$$\int_A^B f(t)dt = \int_a^b f(\varphi(t))\varphi'(t)dt.$$

Proof: Consider the function $G : [a, b] \rightarrow \mathbb{R}$ by

$$G(x) = \int_A^{\varphi(x)} f(t)dt.$$

By the Chain rule, $G'(x) = f(\varphi(x))\varphi'(x)$ for all $x \in [a, b]$. f, φ are cont, so $f\varphi$ is cont and therefore RI. φ' is cont and RI by assumption, so $G'(x) = f(\varphi(x))\varphi'(x)$ is RI. By the Second fundamental theorem,

$$\int_a^b G'(x)dx = G(b) - G(a).$$

$$LHS = \int_a^b f(\varphi(x))\varphi'(x)dx, RHS = \int_A^B f(x)dx - \int_A^A f(x)dx = \int_A^B f(x)dx.$$

Sps $f : [a, b] \rightarrow \mathbb{R}$ is cont, $a < b$. There exists $c \in (a, b)$ st $\int_a^b f(x)dx = (b-a)f(c)$.

Proof: Consider the function $F(x) = \int_a^x f(t)dt$. Since f is cont, F is diff. By MVT, there exists $c \in (a, b)$ st $F'(c) = f(c)$ and $f(c)(b-a) = \int_a^b f(x)dx$.

Let $f : [a, b] \rightarrow \mathbb{R}$ be RI and $g : [a, b] \rightarrow \mathbb{R}$ by

$$g(x) = \begin{cases} f(x) & \text{if } x \neq x_0 \\ y_0 & \text{if } x = x_0. \end{cases}$$

for some $x_0 \in I = [a, b]$. Then f is RI.

Suppose f, g are bdd by M . Let $a = x_0 < x_1 < x_2 < \dots < x_n = b$ be a partition of $I = [a, b]$. Suppose that x_0 is contained in exactly one of the sets $[x_{i-1}, x_i]$ for $i \in [n]$, say for $i = j$. Then

$$\begin{aligned} \sum_{i=1}^n (M_i - m_i)(x_i - x_{i-1}) &= \sum_{i \in [n], i \neq j} (M_i - m_i)(x_i - x_{i-1}) + 2M(x_j - x_{j-1}) \\ &\leq \sum_{i=1}^n (M_i - m_i)(x_i - x_{i-1}) + 2M(x_j - x_{j-1}), \end{aligned}$$

so refine the partition $\{x_i\}$ st $M_i - m_i \rightarrow 0$ (since f is RI) and take $x_j - x_{j-1} \rightarrow 0$.

$\langle e, Ne, N^2e, \dots, N^{m-1}e \rangle$. $Ne + N^{m-1}e$ has height m .

Let V/K be finite dim, $U < V$ st $\alpha|_{U \times U}$ is nondegenerate. Then $V = U \oplus U^\perp$.

Sps $x \in U \cap U^\perp$. Then $\alpha(x, U) = 0 \rightarrow x = 0$, since α is nondegenerate.

I claim $\dim U + \dim U^\perp \geq n$. Let $\{e_1, e_2, \dots, e_m\}$ be a basis of U . $x \in U^\perp$ is equivalent to $\alpha(x, e_i) = 0$ for all $i \in [m]$. Let $f_i = \alpha(\cdot, e_i) \in V^*$. Suppose FTSOC that $\sum \lambda_i f_i = 0$. Then

$$\sum \lambda_i \alpha(x, e_i) = \alpha(x, \sum \lambda_i e_i) = 0$$

In particular this holds for all $x \in U$, so $\sum \lambda_i e_i = 0 \rightarrow \lambda_i = 0$, so the f_i are independent.

Notice $U^\perp = \langle f_1, f_2, \dots, f_m \rangle^\circ$. The LHS has dim $n - m$, so the LHS also has dim $n - m$.

Sps $\alpha(x, x) \neq 0$ for some $x \in V$. Sps $\alpha(x, \lambda x) = 0$. Then $\lambda \alpha(x, x) = 0 \rightarrow \lambda = 0$ since $\alpha(x, x) \neq 0$. Then $\langle x \rangle \oplus \langle x \rangle^\perp = V$. By induction α has the matrix

$$\begin{bmatrix} \lambda_1 & 0 & 0 & \dots & 0 \\ 0 & \lambda_2 & 0 & \dots & 0 \\ 0 & 0 & \lambda_3 & \dots & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & 0 & \dots & \lambda_n \end{bmatrix}$$

Suppose I make the transformation $e_i \mapsto f_i = k_i e_i$. Then the new matrix is

$$\begin{bmatrix} k_1^2 \lambda_1 & 0 & 0 & \dots & 0 \\ 0 & k_2^2 \lambda_2 & 0 & \dots & 0 \\ 0 & 0 & k_3^2 \lambda_3 & \dots & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & 0 & \dots & k_n^2 \lambda_n \end{bmatrix}$$

so if $K = \mathbb{R}$, then each term can be taken to be one of $\{-1, 0, 1\}$.

Sps there exists a basis $\{e_i\}$ of $V/\mathbb{R}$ st $\alpha(e_i, e_i) = 1$ for $i \in [k]$, $\alpha(e_i, e_i) = -1$ for $i \in \{k+1, \dots, k+l\}$, and $\alpha(e_i, e_i) = 0$ otherwise. Suppose $\{f_1, f_2, \dots, f_n\}$ is a basis of n . Then the maximum number of vectors $\{f_i\}$ st $\alpha(f_i, f_i) > 0$ is k .

Write $f_j = \sum_i \lambda_{ij} e_i$. Then $\alpha(f_j, f_j) = \sum_{i=1}^k \lambda_{ij}^2 - \sum_{i=k+1}^{k+l} \lambda_{ij}^2$.